Week 2 | **CLO** 2-C · 3-A · 3-B | **File** 3 / 7

2-2 Histograms, Frequency Polygons and Ogives

المدرج التكراري والمضلع التكراري والمنحنى المتجمع وأشكال التوزيع

Why graphs? Graphs are easier than tables for most people, especially those with little statistical background; they summarise data clearly, and they reveal trends or patterns over time. The workflow is: *organise data → frequency distribution → present → graph*. AMANI

THE THREE GRAPHS OF A GROUPED DISTRIBUTION: WHAT GOES ON EACH AXIS

GRAPH	X-AXIS	Y-AXIS	DRAWN WITH	ENDS
Histogram (frequency or relative frequency) المدرج التكراري	**class boundaries**	f (or RF)	vertical bars that **touch**, equally wide	—
Frequency polygon (or relative frequency polygon) المضلع التكراري	**class midpoints**	f (or RF)	points joined by straight line segments	add one empty class at **each end**, so it touches the x-axis on both sides
Ogive (CF or CRF) المنحنى التكراري المتجمع	**upper class boundaries** (plus the first lower boundary)	CF (or CRF)	points joined by straight line segments	starts at 0 on the **first lower boundary**; ends at n (or 1)

1 · Histogram

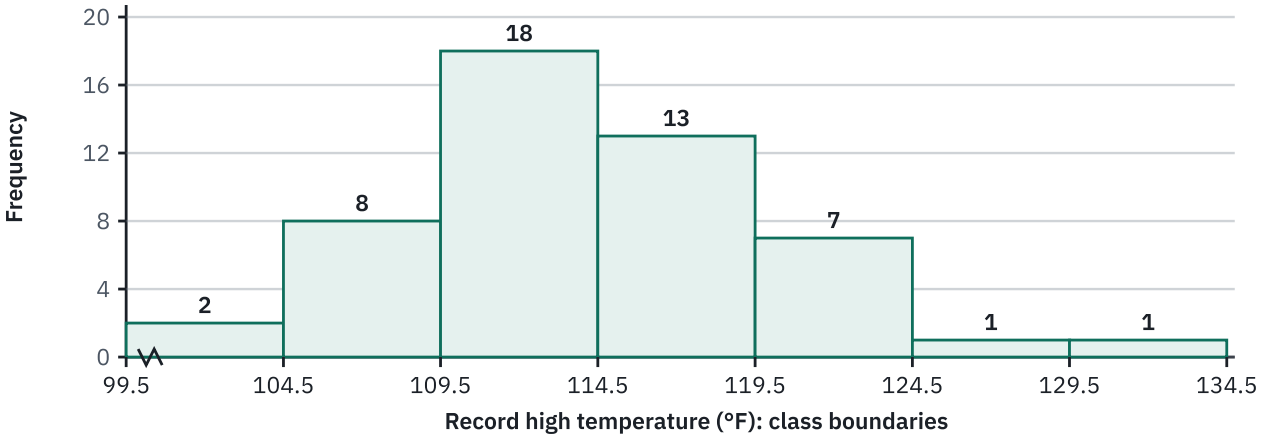
BOTH · CLO2-C

A **histogram** is a **continuous** bar graph showing the number of data items that fall **within** each interval. Bars must touch and be equally spaced.

Steps: (1) draw the x and y axes; (2) put the **class boundaries** on the x-axis and the frequency (or relative frequency) on the y-axis; (3) draw a bar for each class, spanning its boundaries, with height equal to its f (or RF); (4) **title** the graph and **label both axes**.

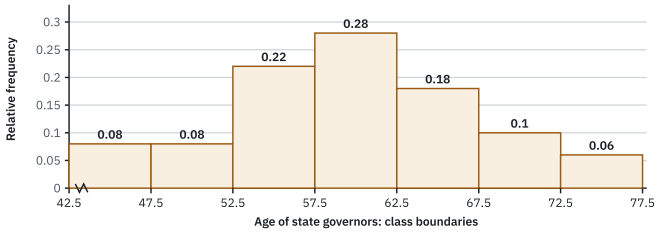
MARKS FOR A CORRECT GRAPH (CLO3-A)

Title · labelled axes with units · boundaries (not limits) under the bars · bars touching · a **jagged / broken line** (zig-zag) on the x-axis when it does not start at 0 · empty classes left as gaps. Histograms are the main tool for judging the **shape** of a distribution (part 4).



Example 2-4. Class boundaries on the x-axis, frequencies on the y-axis, bars touching. The peak is the 109.5–114.5 class (18 states); the distribution is roughly bell-shaped with a slight right tail. The zig-zag marks that the axis does not start at 0.

مثال 2-4: الحدود الفعلية على المحور الأفقي والتكرارات على الرأسي، والأعمدة متلاصقة. القمة في الفئة 109.5–114.5 (18 ولاية)، والشكل قريب من الجرسي مع ذيل خفيف لليمين. الخط المتعرج يعني أن المحور لا يبدأ من الصفر.



Example 2-7. Same shape as a frequency histogram; only the y-axis changes to relative frequency f/n, so the bar heights sum to 1. Use it to compare data sets of different sizes.

مثال 2-7: نفس شكل المدرج التكراري، لكن المحور الرأسي هو التكرار النسبي f/n، ومجموع ارتفاعات الأعمدة = 1. نستخدمه لمقارنة مجموعات مختلفة الحجم.



Dr. Amani's histogram example. The empty 52.5–60.5 class is drawn as a gap with zero height, never removed. Most drivers are 20–52; the two older drivers form a long right tail: right-skewed.

مثال د. أماني: الفئة 52.5–60.5 تكرارها صفر فتظهر فراغًا ولا تُحذف. أغلب السائقين بين 20 و 52، والسائقان الأكبر سنًا يشكلان ذيلًا طويلًا لليمين: التواء موجب.

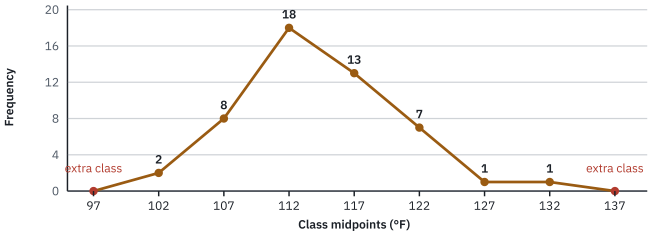
2 · Frequency polygon AHMED'S SLIDES · IN THE OUTLINE

Steps: (1) construct the frequency distribution; (2) draw and label the axes; (3) on the x-axis, mark the **class midpoints** (class marks); (4) plot each class's frequency as a **point**; (5) join neighbouring points with line segments; (6) add one class **below** the lowest and one **above** the highest, each with frequency 0, so the polygon touches the x-axis on both sides.

MIDPOINT (CLASS MARK)

$$X_m = \frac{LCL + UCL}{2} = \frac{LCB + UCB}{2}$$

e.g. (100 + 104)/2 = 102



Example 2-5. Plot frequencies at the class midpoints (not boundaries) and join them with straight lines. Add one empty class at each end (midpoints 97 and 137) so the polygon starts and ends on the x-axis.

مثال 2-5: نرسم التكرارات عند **مراكز** الفئات (لا الحدود) ونصل النقاط بخطوط مستقيمة، ونضيف فئة فارغة في كل طرف (97 و 137) حتى يبدأ المضلع وينتهي على المحور الأفقي.

A **relative frequency polygon** is the same graph with RF on the y-axis. Example 2-7 (governors) has midpoints 45, 50, ..., 75, RF 0.08, 0.08, 0.22, 0.28, 0.18, 0.10, 0.06, and extra end points at 40 and 80.

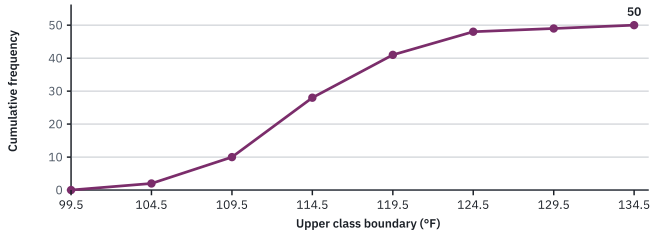
3 · Ogive (cumulative frequency graph) BOTH

An **ogive** is a **line graph** (not bars) of the cumulative frequencies of the classes. **Steps:** (1) draw both axes; (2) on the x-axis mark the first lower boundary and every upper boundary; (3) on the y-axis mark the cumulative frequencies; (4) label the axes; (5) plot each class's CF at its **upper** boundary; (6) join the points with straight lines; (7) start the line at 0 on the first lower boundary; (8) title it.

A **cumulative relative frequency ogive** is identical with CRF on the y-axis, so it ends at 1.

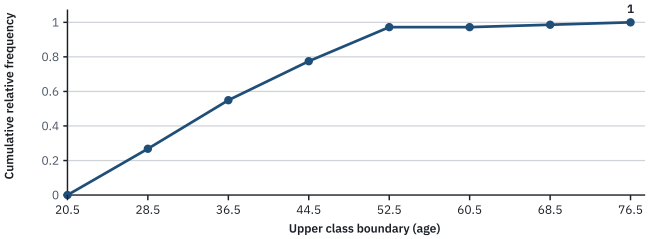
READING AN OGIVE (CLO3-B)

Go up from an x-value to the line, then across: **that many values lie below it**. From the temperature ogive, 41 states had a record below 119.5 °F, so 9 had 119.5 or more. The **steepest** segment is the class with the most data; a **flat** segment is an empty class. Governors (Example 2-7): CF 0, 4, 8, 19, 33, 42, 47, 50; CRF 0, 0.08, 0.16, 0.38, 0.66, 0.84, 0.94, 1.00, so 66% of governors were younger than 62.5.



Example 2-4 ogive. Each point is (upper boundary, cumulative frequency); the line starts at 0 on the first lower boundary. The steepest segment, 109.5–114.5, is the class with the most values. Read it like: 41 states had a record below 119.5 °F.

منحنى التكرار المتجمع: كل نقطة هي (الحد الأعلى، التكرار المتجمع)، ويبدأ من الصفر عند أول حد أدنى. أشد جزء انحدارًا (109.5–114.5) هو الفئة الأكثر تكرارًا. ويُقرأ هكذا: 41 ولاية سجلت أقل من 119.5.



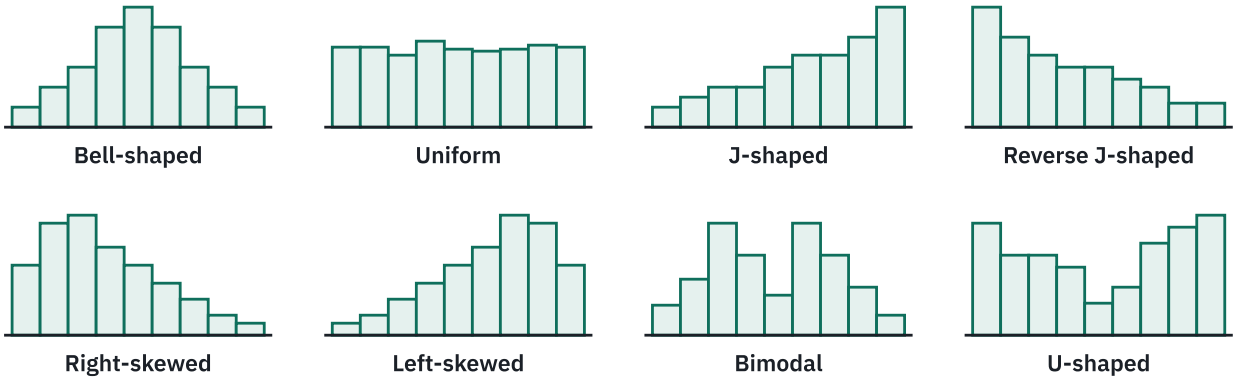
Dr. Amani's second ogive: same points as the cumulative frequency ogive but divided by n = 71, so it always ends at 1. The flat segment from 52.5 to 60.5 is the empty class. About 97% of drivers were younger than 52.5.

المنحنى الثاني لـ د. أماني: نفس النقاط مقسومة على n = 71، فينتهي دائمًا عند 1. الجزء الأفقي (52.5–60.5) هو الفئة الفارغة. حوالي 97% من السائقين أعمارهم أقل من 52.5.

4 · Distribution shapes

BOTH · CLO3-B

Displaying data reveals its shape; the shape tells you the type of distribution, and knowing it is essential for choosing the right analysis (Chapter 3 uses it to choose between the mean and the median).



The eight shapes both decks list. Skew is named after the tail, not the peak: a right-skewed histogram peaks on the left and trails off to the right. Bimodal means two peaks.

الأشكال الثمانية في المحاضرتين. الالتواء يُسمّى باتجاه **الذيل** لا القمة: الملتوي لليمين قمته يسارًا وذيله يمتد لليمين. ثنائي المنوال يعني قمتين.

SHAPE	DESCRIPTION
Bell-shaped	a single peak, values spread evenly on both sides
Uniform	rectangular: about the same frequency everywhere
J-shaped	few values on the left, increasing to the right
Reverse J-shaped	many values on the left, decreasing to the right

SHAPE	DESCRIPTION
Right (positively) skewed	the peak is on the left and most values trail off to the **right** of it
Left (negatively) skewed	the peak is on the right and values trail off to the **left** of it
Bimodal	two peaks of about the same height, with some values between them
U-shaped	high at both ends, low in the middle

الشرح بالعربي — الرسوم الثلاثة للجدول المبوّب

احفظ ما يوضع على المحور الأفقي لكل رسم، فهذا هو السؤال: (1) **المدرج التكراري**: الحدود الفعلية، وأعمدة متلاصقة. (2) **المضلع التكراري**: **مراكز** الفئات ونقاط موصولة، ونضيف فئة فارغة في البداية والنهاية حتى يلمس المحور. (3) **المنحنى المتجمع (Ogive)**: الحدود الفعلية **العليا**، ويبدأ من الصفر عند أول حد فعلي أدنى، وينتهي عند n أو 1. لقراءة المنحنى المتجمع: اصعد من القيمة إلى المنحنى ثم اتجه أفقيًا، فتحصل على عدد القيم الأقل منها. أشد جزء انحدارًا = الفئة الأكثر تكرارًا، والجزء الأفقي = فئة فارغة. وفي الأشكال: الالتواء يُسمّى **باتجاه الذيل**. ولا تنسَ عنوان الرسم وعناوين المحاور، والخط المتعرج إذا لم يبدأ المحور من الصفر.

Chapter 2 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

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